

SPL3 Project Proposal Form, 2026
Institute of Information Technology (IIT)
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Student's Name:	Md. Kamrul Hasan		
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Project Title: Tensor Decomposer Problem Statement and Project Motivation: Modern datasets in fields such as deep learning, image processing, healthcare, computer vision and other complicated problems are usually represented as high-dimensional tensors rather than simple two-dimensional matrices. However, as tensors' dimensionality increases we face many computational challenges. Tensor calculations require heavy computational costs, time and storage. Primary component analysis and singular value decomposition can't fully preserve the multiway relations. Current tensor decomposition implementations are scattered throughout different libraries, difficult to compare and provide no benchmarking. Proposed Solution and Project Description: To address the challenges of analyzing and decomposing higher dimensional tensor data, this project proposes a web based Tensor Decomposer system. The system will allow users to upload or manually input multidimensional tensor data and apply different tensor decomposition algorithms and benchmark results through interactive web interface. The proposed system will have multiple decomposition methods, including, ● CP Decomposition ● Tucker Decomposition ● Higher Order Singular Value Decomposition ● Tensor Train Decomposition The system will provide following functionalities ● Tensor input through web interface ● Tensor decomposition using multiple methods			

- Reconstruction of tensors from decomposed matrices and accuracy analysis
- Compression ratio analysis
- Benchmarking of decomposition method and execution time
- Visualization of decomposed outputs
- Comparison between methods

Supervisor's Name: Mohd. Zulfiqar Hafiz

Signature of the supervisor:

Date: _____

Before the Midterm Presentation 1:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____

Proposal Presentation Feedback (if any) :

Midterm Presentation Feedback (if any) :